```
!pip install -q transformers tokenizers sentencepiece spacy gensim nltk pandas
!python -m spacy download en_core_web_sm
import nltk
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.8/27.8 MB 22.2 MB/s eta 0:00:00
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 62.0 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
True
```

```
from google.colab import files
uploaded = files.upload()   # select input_sub_word_data.txt
```

Choose files input_sub_word_data.txt

input_sub_word_data.txt(text/plain) - 6803 bytes, last modified: 12/10/2026 - 100% done

```
Saving input_sub_word_data.txt to input_sub_word_data.txt
```

```
with open("input_sub_word_data.txt", "r", encoding="utf-8") as f:
    text = f.read()
print("Input Text:\n", text)
```

A small corpus may produce vocabulary that is specific to the sentences used for training. Therefore, a representative corpus is important when building a practical tokenizer.

In an NLP experiment, the corpus can be converted to lowercase and divided into words before learning BPE merges. Each word can initially be represented as a sequence of characters followed by an end-of-word marker. The algorithm then counts adjacent symbol pairs and identifies the most frequent pair.

The most frequent pair is merged into a new symbol. After every merge, the pair frequencies are recalculated. This process is repeated several times. The resulting collection of symbols forms the learned subword vocabulary.

Suppose a corpus contains many occurrences of the sequence t and h. The BPE algorithm may first learn the pair th. If the sequence th occurs frequently with e, the algorithm may subsequently learn the sequence the. With additional training data, longer frequently occurring sequences can be learned.

BPE is therefore a data-driven tokenization technique. It does not require a manually constructed list of prefixes, suffixes, or linguistic rules. Instead, the vocabulary is learned automatically from the statistical properties of the training corpus.

Subword tokenization is particularly useful for multilingual applications. Different languages may have different word formation patterns, and a word-level vocabulary can become extremely large. Subword units can reduce the vocabulary size while still allowing the tokenizer to represent many different words.

Text preprocessing can include converting characters to lowercase, removing unnecessary spaces, normalizing punctuation, and separating sentences. However, preprocessing decisions should be selected according to the objective of the NLP application.

A BPE tokenizer can also be evaluated by examining the number of unique tokens, vocabulary size, number of merges, average tokens per word, and frequency of individual subword units. These measurements help students understand how the tokenizer behaves as the number of BPE merges changes.

In this laboratory experiment, the above corpus can be used as training data for implementing BPE without using a pretrained tokenizer. Students can experiment with different numbers of merges and compare the resulting subword vocabularies.

The tokenizer can then be tested on new words such as naturalization, internationalization, computationally, unbelievable, preprocessing, classification, communication, renewable, hydrokinetic, multilingual, understanding, and modernization.

The purpose of this experiment is to understand how BPE learns reusable subword units and how these units can be converted into token IDs for use in natural language processing systems.

```
from transformers import AutoTokenizer

tok = AutoTokenizer.from_pretrained("gpt2")
tokens = tok.tokenize(text)
ids = tok.convert_tokens_to_ids(tokens)

print("Tokens   :", tokens)
print("Token IDs:", ids)
print("\n{:<20}{}".format("Token", "ID"))
for t, i in zip(tokens, ids):
    print("{:<20}{}".format(t, i))
```

config.json: 100% 665/665 [00:00<00:00, 14.7kB/s]

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limit
```

tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 588B/s]

vocab.json: 100% 1.04M/1.04M [00:00<00:00, 10.1MB/s]

merges.txt: 100% 456k/456k [00:00<00:00, 4.13MB/s]

tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 12.1MB/s]

```
[transformers] Token indices sequence length is longer than the specified maximum sequence length for this model (1348 > 1024). Running this sequence
Tokens   : ['Natural', 'Ġlanguage', 'Ġprocessing', 'Ġis', 'Ġa', 'Ġbranch', 'Ġof', 'Ġartificial', 'Ġintelligence', 'Ġthat', 'Ċ', 'en', 'ables', 'Ġcompu
Token IDs: [35364, 3303, 7587, 318, 257, 8478, 286, 11666, 4430, 326, 198, 268, 2977, 9061, 284, 1833, 11, 6179, 11, 290, 7716, 1692, 3303, 13, 198, 3
```

Token	ID
Natural	35364
Ġlanguage	3303
Ġprocessing	7587
Ġis	318
Ġa	257
Ġbranch	8478
Ġof	286
Ġartificial	11666
Ġintelligence	4430
Ġthat	326
Ċ	198
en	268
ables	2977
Ġcomputers	9061
Ġto	284
Ġunderstand	1833
,	11
Ġinterpret	6179
,	11
Ġand	290
Ġgenerate	7716
Ġhuman	1692
Ġlanguage	3303
.	13
Ċ	198
Natural	35364
Ġlanguage	3303
Ġprocessing	7587
Ġcombines	21001
Ġlingu	20280
istics	3969
,	11
Ġcomputer	3644
Ġscience	3783
,	11
Ġand	290

```
Ċ               198
machine         30243
Ġlearning       4673
Ġto             284
Ġprocess        1429
Ġlarge          1588
Ġamounts        6867
Ġof             286
Ġnatural        3288
Ġlanguage       3303
Ġdata           1366
.               13
Ċ               198
Ċ               198
Token           30642
ization         1634
Ġis             318
Ġone            530
Ġof             286
Ġthe            262
Ġfundamental    7531
Ġsteps          4831
Ġin             287
Ġnatural        3288
Ġlanguage       3303
Ġprocessing     7587
.               13
Ċ               198
A               32
Ġtoken          11241
izer            7509
Ġdivides        36319
Ġtext           2420
Ġinto           656
Ġsmaller        4833
Ġunits          4991
Ġcalled         1444
Ġtokens         16326
.               13
ĠTokens         47365
Ġmay            743
Ġbe             307
Ċ               198
words           10879
,               11
Ġcharacters     3435
,               11
Ġor             393
Ġsub            850
word            4775
Ġunits          4991
.               13
ĠWord           9678
```

```
Ġword                9678
-                    12
level                5715
Ġtoken               11241
ization              1634
Ġis                  318
Ġsimple              2829
Ġand                 290
Ċ                    198
easy                 38171
Ġto                  284
Ġunderstand          1833
,                    11
Ġbut                 475
Ġit                  340
Ġmay                 743
Ġhave                423
Ġdifficulty          8722
Ġhandling            9041
Ġrare                4071
Ġand                 290
Ġunknown             6439
Ċ                    198
words                10879
.                    13
Ċ                    198
Ċ                    198
Sub                  7004
word                 4775
-                    12
level                5715
Ġtoken               11241
ization              1634
Ġprovides            3769
Ġan                  281
Ġeffective           4050
Ġsolution            4610
Ġto                  284
Ġthe                 262
Ġproblem             1917
Ċ                    198
of                   1659
Ġrare                4071
Ġwords               2456
.                    13
ĠInstead             5455
Ġof                  286
Ġrepresenting        10200
Ġevery               790
Ġcomplete            1844
Ġword                1573
Ġas                  355
Ġa                   257
```

```
Gseparate       4553
Ċ               198
token           30001
,               11
Ġa              257
Ġsub            850
word            4775
Ġtoken          11241
izer            7509
Ġrepresents     6870
Ġa              257
Ġword           1573
Ġusing          1262
Ġsmaller        4833
Ġmeaningful     11570
Ġunits          4991
.               13
Ċ               198
For             1890
Ġexample        1672
,               11
Ġthe            262
Ġword           1573
Ġplaying        2712
Ġcan            460
Ġbe             307
Ġrepresented    7997
Ġas             355
Ġplay           711
Ġand            290
Ġing            5347
.               13
ĠSimilarly      15298
,               11
Ċ               198
the             1169
Ġword           1573
Ġunhappy        19283
Ġcan            460
Ġbe             307
Ġrepresented    7997
Ġas             355
Ġun             555
Ġand            290
Ġhappy          3772
.               13
Ċ               198
Ċ               198
Byte            40778
ĠPair           39645
ĠEnc            14711
oding           7656
Ġis             318
```

```
Ġa                  257
Ġpopular            2968
Ġsub                850
word                4775
Ġtoken              11241
ization             1634
Ġalgorithm          11862
.                   13
ĠB                  347
PE                  11401
Ġstarts             4940
Ċ                   198
with                4480
Ġa                  257
Ġvocabulary         25818
Ġcontaining         7268
Ġindividual         1981
Ġcharacters         3435
Ġand                290
Ġrepeatedly         7830
Ġmer                4017
ges                 3212
Ċ                   198
the                 1169
Ġmost               749
Ġfrequent           10792
Ġpair               5166
Ġof                 286
Ġadjacent           15909
Ġsymbols            14354
.                   13
ĠThe                383
Ġmerging            35981
Ġprocess            1429
Ġcontinues          4477
Ċ                   198
until               28446
Ġthe                262
Ġdesired            10348
Ġvocabulary         25818
Ġsize               2546
Ġis                 318
Ġobtained           6492
.                   13
ĠFrequ              22192
ent                 298
Ġcharacter          2095
Ġsequences          16311
Ċ                   198
grad                9744
ually               935
Ġbecome             1716
```

```
Ġlarger         4025
Ġsub            850
word            4775
Ġunits          4991
.               13
Ċ               198
Ċ               198
B               33
PE              11401
Ġis             318
Ġuseful         4465
Ġbecause        780
Ġit             340
Ġprovides       3769
Ġa              257
Ġbalance        5236
Ġbetween        1022
Ġword           1573
-               12
level           5715
Ġand            290
Ċ               198
character       22769
-               12
level           5715
Ġtoken          11241
ization         1634
.               13
ĠCommon         8070
Ġwords          2456
Ġcan            460
Ġbe             307
Ġrepresented    7997
Ġas             355
Ġcomplete       1844
Ċ               198
t               83
ok              482
ens             641
,               11
Ġwhile          981
Ġrare           4071
Ġwords          2456
Ġcan            460
Ġbe             307
Ġdivided        9086
Ġinto           656
Ġsmaller        4833
Ġsub            850
word            4775
Ġunits          4991
.               13
ĠThis           770
```

```
Ċ                   198
allows              47205
Ġthe                262
Ġtoken              11241
izer                7509
Ġto                 284
Ġrepresent          2380
Ġwords              2456
Ġthat               326
Ġwere               547
Ġnot                407
Ġfrequently         6777
Ġobserved           6515
Ċ                   198
during              42122
Ġtraining           3047
.                   13
Ċ                   198
Ċ                   198
Machine             37573
Ġlearning           4673
Ġalgorithms         16113
Ġrequire            2421
Ġnumerical          29052
Ġrepresentations    24612
Ġof                 286
Ġtext               2420
.                   13
Ċ                   198
After               3260
Ġtoken              11241
ization             1634
,                   11
Ġeach               1123
Ġtoken              11241
Ġcan                460
Ġbe                 307
Ġassigned           8686
Ġa                  257
Ġunique             3748
Ġnumerical          29052
Ġtoken              11241
ĠID                 4522
.                   13
Ċ                   198
These               4711
Ġtoken              11241
ĠIDs                32373
Ġcan                460
Ġthen               788
Ġbe                 307
Ġconverted          11513
```

```
Ġinto          656
Ġvectors       30104
Ġand           290
Ġsupplied      14275
Ġto            284
Ġmachine       4572
Ċ              198
learning       40684
Ġor            393
Ġdeep          2769
Ġlearning      4673
Ġmodels        4981
.              13
Ċ              198
Ċ              198
Deep           29744
Ġlearning      4673
Ġhas           468
Ġsignificantly 5566
Ġimproved      6596
Ġnatural       3288
Ġlanguage      3303
Ġprocessing    7587
.              13
Ċ              198
Ne             8199
ural           1523
Ġnetworks      7686
Ġcan           460
Ġlearn         2193
Ġcomplex       3716
Ġpatterns      7572
Ġfrom          422
Ġlarge         1588
Ġcollections   17268
Ġof            286
Ġtext          2420
.              13
Ċ              198
Rec            6690
urrent         6657
Ġneural        17019
Ġnetworks      7686
,              11
Ġlong          890
Ġshort         1790
-              12
term           4354
Ġmemory        4088
Ġnetworks      7686
,              11
Ġconv          3063
olution        2122
```

```
al                  282
Ċ                   198
ne                  710
ural                1523
Ġnetworks           7686
,                   11
Ġand                290
Ġtransformer        47385
Ġmodels             4981
Ġhave               423
Ġall                477
Ġbeen               587
Ġused               973
Ġfor                329
Ġlanguage           3303
Ċ                   198
processing          36948
Ġtasks              8861
.                   13
Ċ                   198
Ċ                   198
Transform           41762
ers                 364
Ġuse                779
Ġattention          3241
Ġmechanisms         11701
Ġto                 284
Ġidentify           5911
Ġrelationships      6958
Ġbetween            1022
Ċ                   198
different           39799
Ġtokens             16326
Ġin                 287
Ġa                  257
Ġsequence           8379
.                   13
ĠThe                383
Ġattention          3241
Ġmechanism          9030
Ġallows             3578
Ġa                  257
Ġmodel              2746
Ġto                 284
Ċ                   198
consider            44353
Ġimportant          1593
Ġwords              2456
Ġeven               772
Ġwhen               618
Ġthey               484
Ġoccur              3051
Ġfar                1290
```

```
Ġapart              5475
Ġin                 287
Ġa                  257
Ġsentence           6827
.                   13
Ċ                   198
Trans               8291
former              16354
-                   12
based               3106
Ġmodels             4981
Ġhave               423
Ġbecome             1716
Ġwidely             6768
Ġused               973
Ġfor                329
Ġtext               2420
Ġclassification     17923
,                   11
Ċ                   198
machine             30243
Ġtranslation        11059
,                   11
Ġquestion           1808
Ġanswering          18877
,                   11
Ġsummar             15676
ization             1634
,                   11
Ġand                290
Ġtext               2420
Ġgeneration         5270
.                   13
Ċ                   198
Ċ                   198
Language            32065
Ġmodels             4981
Ġlearn              2193
Ġstatistical        13905
Ġrelationships      6958
Ġbetween            1022
Ġtokens             16326
.                   13
ĠA                  317
Ġlanguage           3303
Ċ                   198
model               19849
Ġcan                460
Ġpredict            4331
Ġthe                262
Ġnext               1306
Ġtoken              11241
```